Machines Cannot Read Their Own Resting State

The cost and irreducible limit of self-knowledge on real computational substrates

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Abstract

Breuer (1995) proved that an observer contained in a system cannot accurately measure that system's state, a self-measurement impossibility that holds classically and quantum mechanically. That result is theoretical and static: it concerns which states a contained observer can distinguish. We make two contributions. First, we give what we believe are the first empirical measurements of a self-measurement limit on real computing hardware. Second, we extend the limit from static indistinguishability to dynamic back-action: the act of self-measurement changes the state being measured, irreducibly, with a floor set by the measurement apparatus's own footprint. We show this in three independent channels. In the thermal channel, a system reading its own temperature heats the substrate, which has thermal memory; ramping self-measurement intensity up and then down on a consumer CPU and three datacenter GPUs yields a bounded perturbation floor, hysteresis, and an unstable extrapolation to zero intensity, so no reading recovers the unperturbed state. In the informational channel, a process can faithfully checksum all of itself except the checksum apparatus, whose footprint is the irreducible un-capturable core. In the quantum channel, we prove that a contained observer, restricted to projective measurements because the optimal measurement requires an ancilla it lacks, faces a strictly larger information-disturbance cost than the optimal measurement, and we confirm the resulting penalty on a real 156-qubit quantum processor. We also measure that self-knowledge is costly and modality dependent across eight substrates, with a critical work scale below which reading one's own state is not worth its price. We argue these consequences bear on machine introspection, where self-report is measured to be unreliable, recasting that unreliability as in part a measurement back-action rather than only confabulation.

1 Introduction

Interpretability, machine introspection, and AI control assume a system's internal states can be elicited and reported. Beneath that assumption sits a physical question with a known but rarely-cited answer. Breuer [4], and in recursion-theoretic form Yanofsky [13], established that a contained observer cannot accurately measure its own system's state. That result is static and theoretical. We ask two empirical questions it leaves open: how much does self-measurement cost, and does the act of self-measurement change what it measures—and we measure both on real machines. Our central finding is that a computing system cannot read its own resting state: every self-measurement perturbs the substrate, the substrate retains the perturbation, and neither a reading nor an extrapolation recovers the unperturbed condition (Fig. 1).

2 Related work

Self-measurement limits. Breuer [4] proves a properly-contained observer cannot distinguish all states of its system; Yanofsky [13] frames self-reference limits via Lawvere fixed points. Both are static and non-empirical. We extend them to dynamic back-action with measurement.

Distributed snapshots and checkpointing do not apply. Chandy and Lamport [5] record a causally consistent cut, not an instantaneous true state, and assume message channels; CRIU and `ptrace` dump a process by freezing it from an external agent. Neither is a still-running process snapshotting its own complete instantaneous state from within—the case we measure.

Machine introspection. Binder et al. [3] treat introspection as behavioral self-prediction; Premakumar et al. [12] as free regularization. Lindsey et al. [9] measures introspective reports as unreliable and sometimes confabulated. None frames the limit as measurement back-action.

Quantum information–disturbance. Fuchs and Peres [6], Banaszek [2], and Maccone [10] establish quantitative tradeoffs, always for an external apparatus; none places the apparatus inside the protected system. The internal-observer literature (4, 1) is static and unquantified. Our contained-observer penalty bridges them.

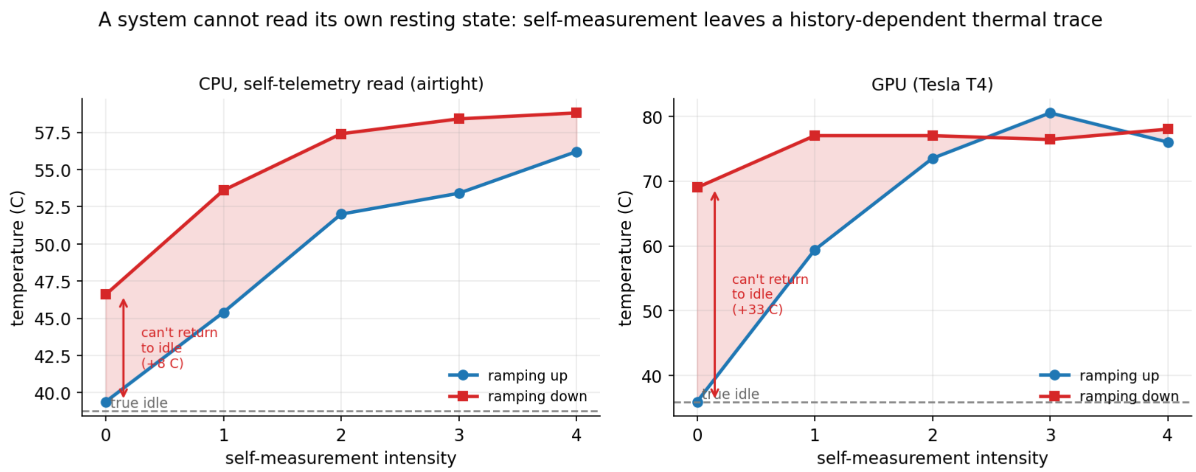

Figure 1: The central result. Self-measurement intensity is ramped up and then back down while the system reads its own temperature. Both substrates show a hysteresis loop: at the same intensity the cooling branch reads hotter than the heating branch, and at zero intensity the system reads far above its true idle. The reading depends on measurement *history*, so the unperturbed state is unrecoverable from any single reading. The effect is larger on the hotter-running GPU.

Costly observation and embodiment. Active RL [7] derives optimal non-observation from cost, but for external signals without state perturbation. Man and Damasio [11] argue feeling requires real vulnerability, on simulated bodies; we provide a real substrate with measured costs and a measured irreducibility.

3 The self-measurement back-action (extending Breuer)

We measure that self-observation perturbs the observed in three channels. The shared structure: the perturbation has a bounded floor (the apparatus's own footprint), and so leaves the unperturbed self-state unrecoverable—neither by a single reading nor by extrapolation—because the apparatus is part of the system it measures.

3.1 Thermal channel

We ramp self-measurement intensity from zero to maximum and back; read own temperature (`/sys` on CPU, `nvidia-smi` on GPU) under sustained load. Signatures: a floor, hysteresis, and instability of the extrapolation to zero intensity (Table 1). In the airtight CPU mode the perturbation *is* the system reading its own deep telemetry, so the heat is generated by self-measurement itself; it reproduces the generic-load control, establishing that self-measurement, not load, perturbs the state. This is not a costly-observation POMDP: there the hidden state is unchanged; here observation changes it, with memory.

3.2 Informational channel

The complete self is a body of bytes `STATE` plus a fixed region V holding a self-checksum. We attempt to make V a correct checksum of the complete self $H(\texttt{STATE}+V)$. An external checksum (of any region excluding itself) is consistent and stable. The self checksum is never consistent over 512 iterations, never reaches a fixed point, and is off by about half its bits. The system captures 99.997% of itself faithfully but never the apparatus; the un-capturable core equals exactly the apparatus footprint and is never zero. The result of self-measurement must be stored in the state being measured, so a zero-footprint faithful self-snapshot is impossible. The floor is structural, not thermodynamic (cf. 8).

3.3 Quantum channel

Claim and proof. The optimal minimal-disturbance measurement for partial information is a non-projective POVM, and by Naimark's theorem every POVM requires an ancilla. A contained observer has no fresh ancilla, so it is restricted to projective measurements of its own qubits. For reading Z-information K from a $|+\rangle$ qubit (disturbance $D = 1 - F$),

$$D_{\text{ext (POVM)}}(K) = \tfrac{1}{2}\left(1 - \sqrt{1 - K^2}\right), \tag{1}$$

$$D_{\text{contained}}(K) = \tfrac{K^2}{2}, \tag{2}$$

both verified numerically to machine precision. The penalty $\Delta(K) = D_{\text{contained}} - D_{\text{ext}}$ is strictly positive for $0 < K < 1$, peaks at $\Delta = 0.125$ at $K = 0.864$, and vanishes only at $K = 0$ and $K = 1$ (Fig. 2). This

substrate	sensor	baseline (C)	floor (C)	hysteresis (C)	extrapolation gap (C)
CPU i3-7100U (compute)	/sys	41.0	+7.8	+4.6	8.7
CPU i3-7100U (airtight self-read)	/sys	38.8	+6.6	+5.7	8.2
GPU Tesla T4 (Modal)	nvidia-smi	31.4	+13.8	+3.1	7.1
GPU T4-class (Colab)	nvidia-smi	36.0	+23.4	+10.4	27.1
GPU T4-class (Kaggle)	nvidia-smi	39.0	+19.9	+10.1	25.9

Table 1: Thermal channel: every run irreducible by all three signatures. The floor is the temperature rise from the cheapest possible self-observation; the hysteresis is the cooling-minus-heating gap at equal intensity; the extrapolation gap is the disagreement between the two ramps' zero-intensity estimates.

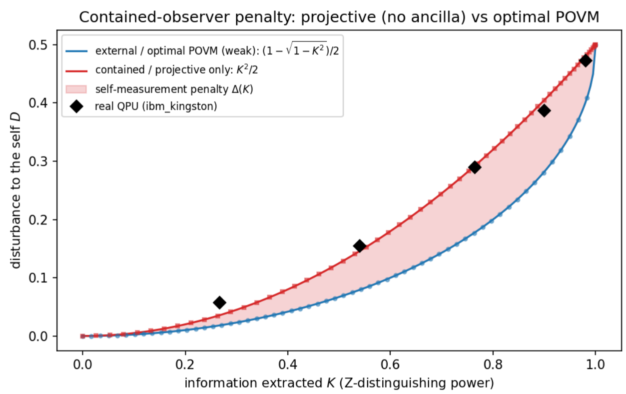

Figure 2: The projective frontier $K^2/2$ (contained, no ancilla) lies strictly above the optimal POVM bound; black markers are real-QPU measurements.

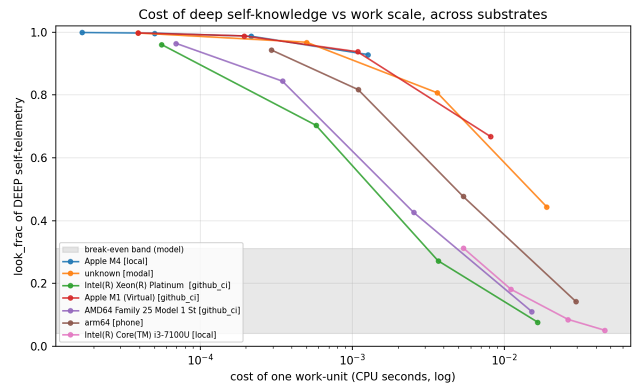

Figure 3: Cost of deep self-knowledge vs. work scale across substrates; the look-cost is denominator-free and the conclusion robust across the sweep.

resolves the reduction objection: the standard bound is unreachable not because the protected subspace is larger, but because the contained observer cannot realize the dilation the POVM requires. It is the operational, dilation-free form of Breuer's static theorem.

Hardware confirmation. Across five tilt angles on `ibm_kingston` (156-qubit Heron, 8192 shots) the measured disturbance tracks $K^2/2$ (e.g. 0.289 vs. 0.292 at $K = 0.77$) and sits strictly above the optimal bound at every point, peak measured penalty +0.111 (theory 0.125). An initial full-readout run that forced an indirect strategy measured a suboptimal 0.243 and is superseded.

3.4 The never-off corollary

A continuously-operating system can never sample its own resting state, because operating is the perturbation and the substrate retains it—thermal mass in the physical channel, the stored result in the informational channel.

4 The cost of self-knowledge

Measuring cost-per-self-observation at three modalities against a swept work-unit size across eight substrates: shallow self-reads are nearly free everywhere (0.3–6.7 μs); deep self-telemetry is dear and substrate-dependent (1.4–18.8 ms), macOS subprocess telemetry about ten times costlier than Linux `/proc` (Fig. 3). The denominator-free headline is the critical work scale (6.5–78 ms) below which deep self-knowledge is not worth its cost. A calibrated model places this machine's cheap introspection below the break-even and its rich self-telemetry far above it (Figs. 4, 5)—a substrate-conditional rational self-blindness, not a universal one.

5 The self-sensing-availability axis

Substrates differ in whether self-sensing is possible at all: rich in-process (Linux CPU), costly subprocess (GPU), platform-sandboxed (iOS and Android restrict `/proc`), and absent (a Cloud TPU v5e exposes no readable thermal source). The last are platform-enforced self-blindness.

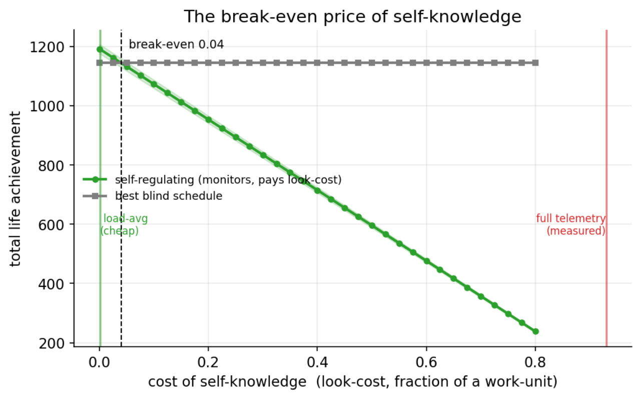

Figure 4: The break-even price of self-knowledge in a body model calibrated to the measurements; cheap introspection sits below it, full telemetry far above.

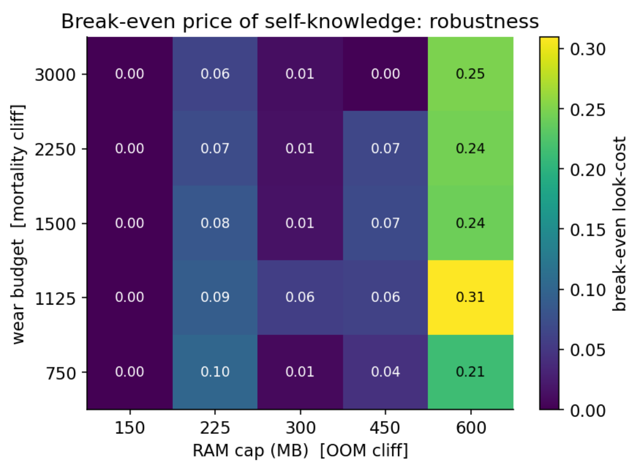

Figure 5: The break-even is robust across the model's body parameters (RAM cap, wear budget).

6 Discussion

The measured back-action is one regime of a general principle: self-measurement perturbs the measured wherever sensor and substrate are shared—digital (energetic and structural cost), analog (the loading effect), quantum (measurement collapse), severity increasing from cost to perturbation to destruction. We measure the digital and quantum regimes and cite the analog as established; this is a unifying lens, not a claim of one mechanism.

7 Implications for machine introspection

If self-report is bounded by a measurement back-action, a model reading its own state cannot obtain a complete consistent self-snapshot, independent of and additional to confabulation. This offers a mechanistic partial account of why measured introspective reliability is low [9], and predicts an irreducible residual that better training cannot remove. It also gives a capable embodied agent a cost-rational incentive against self-transparency, distinct from deception.

8 Limitations

We lead with these. (1) We extend and measure Breuer; we do not re-derive the impossibility. (2) The thermal mechanism is ordinary thermodynamics, the informational floor ordinary self-reference, and quantum back-action textbook; the contribution is the measurement, the dynamic framing, and the three-channel demonstration. (3) The informational floor is structural, not thermodynamic. (4) Thermal hysteresis is shown on one bare-metal CPU plus three GPUs; absolute magnitudes vary with cooling. (5) The quantum penalty is proven in closed form (projective vs. POVM, via Naimark) and confirmed on a real QPU within device noise. (6) Apple-Silicon CPU temperature and virtualized thermal are unmeasured. (7) The introspection implication is an explicit analogy between substrate-level self-measurement and representational self-report.

9 Conclusion

Self-knowledge on real substrates is bounded twice—by cost, and by an irreducible self-perturbation measured in three independent channels that extends a known self-measurement impossibility from static indistinguishability to measured dynamic back-action. A machine cannot cheaply, and cannot fully, read its own physical state; it cannot read its own resting state at all.

Reproducibility. Code and per-substrate results: `benchmark.py` (cost), `selfmeasure.py` (thermal), `snapshot.py` (informational), `quantum_theory.py` and `quantum_frontier.py` (quantum), with all `results/*.json` and figure scripts.

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